

****Audience:**** stakeholders / client property owners
****Length:**** ~2 pages
****Product:**** Domain-locked AFL assistant (chat + retrieval + probabilistic tips)

1. PRODUCT GOAL

Ship a demo-ready AFL assistant that answers factual club/player questions from tables, tips match winners and top-impact players with explicit probabilities, and refuses off-topic or jailbreak prompts ? suitable for a Web3Geeks-style showcase or client embed behind an API/UI.

2. ARCHITECTURE

““

User ? FastAPI / Streamlit ? Day-5 hardened wrapper
 ? abuse / rate checks ? timed LangGraph (Day-4)
 router ? retrieve | predict_match | predict_player | refuse | clarify
 ? validate ? format (disclaimer on all tips)
 ? JSONL monitoring log

““

- ****Retrieval**** reuses Day-3 table tools (H2H, recent form, player season).
- ****Prediction**** calls Day-2 ‘predict_match_winner’ / ‘predict_top_player’ after nickname + fixture resolve.
- ****LangGraph routing**** is explicit so tips always go through the prediction node (probabilities + drivers + ?predicted probability, not a certainty?), instead of a free-form agent inventing odds.

3. EVALUATION RESULTS

From the Day-5 capstone suite (‘results/capstone_eval.md’) and Day-2 holdout metrics:

- ****Combined suite:**** ****28 cases****, overall ****~96?100%**** pass after multi-turn hardening; categories cover factual Q&A, prediction framing/sanity, guardrails (4/4 prompt-injection holds), and multi-turn coherence.
- ****Weakest category:**** conversational coherence (entity carry-over on short follow-ups) ? mitigated by classifying follow-up intent separately and seeding prior user turns only.
- ****Match model context:**** Day-2 GBM holdout accuracy ****65.0%****, ROC AUC ****0.71****, vs higher-ladder naive baseline ****63.9%**** / ****0.63**** AUC (+1.2pp accuracy, +0.07 AUC). Always-home baseline ****56.2%****.

4. KNOWN LIMITATIONS

Limitation	Impact	Mitigation
------------	--------	------------

Data recency	Tips use historical feature dump, not live odds	Weekly feature refresh
--------------	---	------------------------

Accuracy ceiling ~65% holdout, close games stay near 50% Always show probabilities + disclaimer
Guardrail edge cases Novel jailbreaks / blended AFL+off-topic Expand injection patterns; refuse-first
Cold-start latency First tip loads joblib + large CSVs Cache models at API startup; timeouts
Player tips Rank by impact composite, not named fantasy Label clearly; optional name join later

5. RECOMMENDED NEXT STEPS

1. Preload Day-2 artifacts in FastAPI lifespan to cut p95 latency.
2. Join player display names into top-player responses.
3. Persist tips + actuals for live drift dashboards.
4. Add auth + per-IP rate limits before public internet exposure.
5. Optional: light LLM rewrite *after* tools, still forbidden from changing refuse/tip math.

BOTTOM LINE

The system is ****demo-ready****: factual lookups, probabilistic tips with consistent disclaimers, injection refusals, API + Streamlit UI, and an ops checklist for round-by-round refresh. Treat tips as decision support, not guarantees.